

From BERT to Frontier Agents: Eight Years of Language-Model Progress, the Collapse of the Capability–Cost Curve, and the Rise of Task-Targeted Models

An empirical review with reproducible measurements, 2018–2026

August 2026

Abstract

Between October 2018 and July 2026, language models moved from BERT, a 340-million-parameter encoder that could barely generate text, to trillion-parameter agentic systems that solve International Mathematical Olympiad problems, repair real software repositories, and operate desktop computers. This paper reviews that progression, quantifies it, and tests three practical claims with code. First, using public benchmark series (GLUE, SuperGLUE, MMLU, SWE-bench Verified), we estimate that agentic-coding capability, measured as the odds of solving a real GitHub issue, has grown by a factor of $\sim 5.8\times$ per year since October 2024. Second, we document the collapse of the capability–cost curve: GPT-5.6 Luna, OpenAI’s budget tier at \$1/\$6 per million input/output tokens, matches or beats GPT-5.5 — the flagship of eleven weeks earlier — on most agentic and professional-work benchmarks selected from the vendor’s own tables at one-fifth the token price, and exceeds the August 2025 flagship GPT-5’s SWE-bench Verified score by ~ 18 points on an independent harness’s reconstruction. Third, we show that the frontier has fragmented by task: Claude Opus 5 leads human-preference frontend coding, Claude Fable 5 leads repository-level coding, GPT-5.6 Sol leads agentic terminal work, and Opus 5 quadrupled the record on the ARC-AGI-3 novel-reasoning benchmark. An oracle that routes each task to the best of six public models recovers the full per-benchmark optimum, and we show a two-model router already captures all of that gain. Finally, as a proof of method, we run a fully traced 16-item pilot with a 0.5-billion-parameter open model on grade-school math. We compare greedy decoding with self-consistency sampling, report paired uncertainty, and demonstrate a reproducible way to measure test-time-compute interventions without retraining. All data, code, unit tests, and compilation scripts are provided.

Keywords: large language models; evaluation; scaling; inference economics; task specialization; self-consistency; model routing

Contents

1	Introduction	3
2	Eight years in four paradigms	3
2.1	The encoder era: BERT and the fine-tuning paradigm (2018–2019)	3
2.2	Scale and in-context learning: GPT-3 (2020–2021)	4

2.3	Alignment and instruction following (2022–2023)	4
2.4	Reasoning and agents (2024–2026)	4
3	Data and methods	4
3.1	Data	4
3.2	Analysis methods	6
4	Results	6
4.1	Capability growth: three benchmark eras	6
4.2	The collapse of the capability–cost curve	7
4.3	The frontier fragmented by task	7
5	Getting more from existing models: three suggestions, tested	9
5.1	Suggestion 1: Spend test-time compute; sample and vote	9
5.2	Suggestion 2: Calibrate reasoning effort; more is not better	10
5.3	Suggestion 3: Route tasks to specialized models	10
5.4	What we would tell the labs	10
6	Discussion: how models get good at specific tasks	11
7	Conclusion	12

1 Introduction

In October 2018, Google released BERT, a bidirectional Transformer encoder that set the state of the art on eleven natural-language understanding tasks and pushed the GLUE benchmark to 80.5%¹. BERT could not write a coherent paragraph; it was a *reading* model, fine-tuned per task. In July 2026, Anthropic released Claude Opus 5 and reported 42/42 under its post-contest, no-tools grading protocol for the 2026 International Mathematical Olympiad problems; ARC Prize independently measured a roughly fourfold jump on ARC-AGI-3^{2,3}. Between those two releases sit eight years in which the field’s dominant paradigm changed at least four times: from task-specific fine-tuning, to few-shot prompting of large pretrained decoders, to instruction tuning and human-feedback alignment, to reasoning models and tool-using agents.

This paper has three goals. The first is *descriptive*: to trace the technical progression from BERT to the 2026 frontier — GPT-5.6 (Sol, Terra, Luna), Claude Fable 5 and Opus 5, Kimi K3, and their contemporaries — in one place, with numbers. The second is *quantitative*: to measure how fast capability and cost-efficiency have actually improved, using public benchmark and pricing series rather than impressions. The third is *prescriptive*: to test, with runnable code, three practical suggestions for getting more out of existing models — spending inference compute better, calibrating reasoning effort, and routing tasks across models.

Contributions.

1. A consolidated, sourced timeline of language-model capability from 2018 to 2026 across three benchmark eras (Section 2, Table 1).
2. Trend fits showing $\sim 5.8\times$ annual growth in the odds of solving SWE-bench Verified tasks, alongside the saturation of knowledge benchmarks such as MMLU (Section 4.1).
3. A cost analysis showing a $\sim 60\times$ decline in input-token prices from GPT-3 (2020) to GPT-5.6 Luna (2026), and a demonstration that today’s budget tier matches the flagship of one quarter ago on most agentic/professional benchmarks (Section 4.2).
4. Evidence that the 2026 frontier is task-fragmented, and that simple model routing recovers the per-task optimum (Section 4.3).
5. A live, reproducible 16-item pilot measuring self-consistency sampling against greedy decoding for a small open model, with complete traces and paired uncertainty, plus analyses of reasoning-effort calibration (Section 5).

2 Eight years in four paradigms

2.1 The encoder era: BERT and the fine-tuning paradigm (2018–2019)

BERT¹ established the pretrain-then-fine-tune recipe: masked-language-model pretraining on unlabeled text, followed by a small task-specific head. RoBERTa⁴ showed the recipe was undertrained rather than wrong, reaching 88.5 on GLUE with more data and longer training; T5⁵ unified all tasks as text-to-text and reached 90.3 on SuperGLUE. GPT-2⁶ demonstrated that a left-to-right decoder trained on enough web text acquired tasks *without* fine-tuning, but its zero-shot quality was not yet competitive. The era’s signature: capability lived in task-specific heads, and benchmarks (GLUE, SuperGLUE, SQuAD) saturated within roughly two years of introduction.

2.2 Scale and in-context learning: GPT-3 (2020–2021)

GPT-3⁷ scaled the decoder recipe to 175B parameters and showed that few-shot prompting — a handful of examples in the context window — could substitute for gradient updates on many tasks. Its 43.9% five-shot MMLU⁸ looks modest today, but the conceptual shift was permanent: capability could be *elicited* from a frozen general model. The limits were equally clear: GPT-3 few-shot scored 71.8 on SuperGLUE, well below fine-tuned systems, and its unaligned samples were unreliable.

2.3 Alignment and instruction following (2022–2023)

InstructGPT⁹ applied reinforcement learning from human feedback (RLHF) to make models follow instructions; ChatGPT (November 2022) packaged this for general users. GPT-4 (March 2023) jumped to 86.4% on MMLU¹⁰, near the estimated human-expert level, and established the modern pattern of a capability report accompanying release. Meta’s Llama series opened weights and created an open ecosystem; Claude and Gemini established a multi-vendor frontier.

2.4 Reasoning and agents (2024–2026)

Two further shifts followed. First, *test-time compute*: OpenAI’s o1 (December 2024) showed that training models to produce long chains of thought — and spending more inference tokens — improved math and coding substantially; o1 reached 91.8% on MMLU, effectively saturating it¹¹. Second, *agency*: SWE-bench Verified^{12,13} made “resolve a real GitHub issue” the standard hard task, and scores climbed from 49% (Claude 3.5 Sonnet, October 2024) to 97% (Claude Opus 5, July 2026) on independent harnesses¹⁴. By mid-2026 the flagship releases — Claude Fable 5 (July 1), GPT-5.6 Sol/Terra/Luna (July 9), Kimi K3 (July 16), Claude Opus 5 (July 24) — were evaluated primarily on agentic, long-horizon, and economically-weighted tasks rather than static question answering^{15–18}. Kimi K3 additionally marked the near-closure of the open–closed gap: a 2.8-trillion-parameter open-weight mixture-of-experts model competitive with the best proprietary systems¹⁷.

Table 1 summarizes the arc, and Figure 2 shows the agentic benchmark series at its current endpoint.

3 Data and methods

3.1 Data

We assembled three analysis tables plus one evaluation split, all shipped with this paper’s code.

D1: Timeline. Twenty-seven (model, benchmark, score, date, source) rows spanning 2018–2026, drawn from the original papers and from leaderboard mirrors (Table 1; full file `data/models_timeline.csv`). Where multiple evaluation variants exist we record the variant explicitly (e.g., vendor-reported vs. independent SWE-bench Verified).

D2: 2026 frontier matrix. Per-model benchmark scores and API prices for the July 2026 frontier — GPT-5.6 Sol/Terra/Luna, Claude Fable 5, Claude Opus 5, Kimi K3, plus the April 2026 flagship GPT-5.5 and Claude Opus 4.8 — from OpenAI’s GPT-5.6 launch tables¹⁶, Anthropic’s Opus 5 launch and system card^{2,18}, Artificial Analysis^{19,20}, the Vals AI independent SWE-bench Verified harness¹⁴, and arena.ai’s human-preference Frontend Code Arena²¹.

Table 1: Selected milestones, 2018–2026. Scores are as originally reported by the cited source; benchmark variants are noted.

Date	Model	Org	Milestone metric	Score
2018-10	BERT-Large	Google	GLUE	80.5
2019-07	RoBERTa	Meta	GLUE	88.5
2019-10	T5-11B	Google	SuperGLUE	90.3
2020-05	GPT-3	OpenAI	MMLU (5-shot)	43.9
2023-03	GPT-4	OpenAI	MMLU (5-shot)	86.4
2024-05	GPT-4o	OpenAI	MMLU	88.7
2024-12	o1	OpenAI	MMLU	91.8
2024-10	Claude 3.5 Sonnet	Anthropic	SWE-bench Ver.	49.0
2025-05	Claude Opus 4	Anthropic	SWE-bench Ver.	72.5
2025-08	GPT-5	OpenAI	SWE-bench Ver.	74.9
2025-11	Claude Opus 4.5	Anthropic	SWE-bench Ver.	76.8
2026-04	GPT-5.5	OpenAI	SWE-bench Ver. (vendor)	85.1
2026-07	Claude Fable 5	Anthropic	SWE-bench Ver. (indep.)	95.0
2026-07	GPT-5.6 Sol	OpenAI	SWE-bench Ver. (indep.)	96.2
2026-07	Kimi K3	Moonsot	SWE-bench Ver. (indep.)	93.4
2026-07	Claude Opus 5	Anthropic	SWE-bench Ver. (indep.)	97.0

Reproducibility architecture

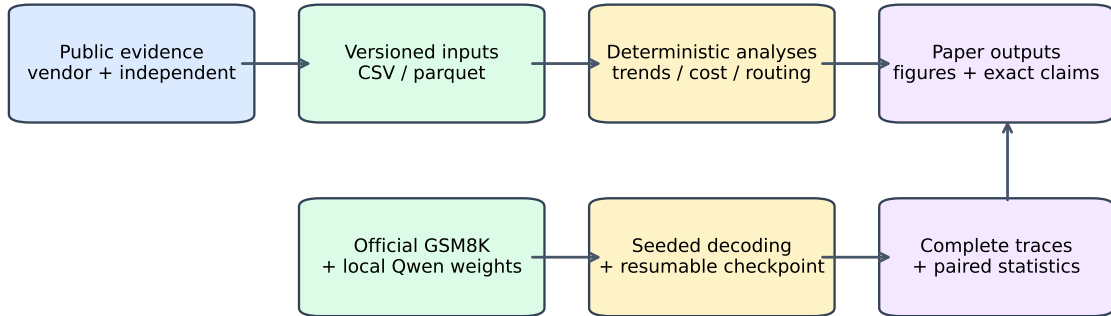


Figure 1: Reproducibility architecture. Public evidence and versioned inputs feed deterministic analyses; the live experiment separately records complete per-problem generations and paired statistics before either path reaches the paper.

D3: Pricing. Input/output prices per million tokens for representative API models from 2020 to 2026, using vendor documentation where available^{16,22}.

D4: Live-evaluation items. The committed official GSM8K test split contains 1,319 question/answer pairs²³; the live pilot uses a fixed 16-item sample and records the original row identifiers.

Provenance caveat. Mid-2026 numbers mix vendor-reported and independent evaluations. We label provenance wherever it matters and prefer third-party measurements (Vals AI, Artificial Analysis, ARC Prize) when both exist. Vendor launch numbers are claims, not independent results; we treat them as such and note where vendor and third-party harnesses differ.

3.2 Analysis methods

Trend fitting. For the SWE-bench Verified series we fit a log-linear model to the *odds* of solving a task, $\text{logit}(p)$, against release date, and report the implied annual multiplicative growth of those odds with the fit R^2 (Figure 2). For MMLU we compare the raw series with the estimated human-expert level of 89.8%⁸.

Pareto analysis. Using Artificial Analysis’s all-effort dataset for the GPT-5.6 family (15 model×effort settings with intelligence-index scores and per-task costs^{19,20}), we compute the cost-capability Pareto frontier and identify dominated settings.

Routing oracle. On the fourteen benchmarks that OpenAI’s GPT-5.6 launch table reports for all six of Sol, Terra, Luna, GPT-5.5, Fable 5, and Opus 4.8, we normalize each benchmark to its best model (100), then compare (i) the best single model’s mean, (ii) an oracle that picks the best model per benchmark, and (iii) restricted routers (a cheapest-within-one-point router and a two-model router).

Live experiment. Section 5.1 describes a controlled inference-time pilot on Qwen2.5-0.5B-Instruct. We select 16 items from the official 1,319-item GSM8K test split with data seed 42 and compare deterministic greedy decoding with eight sampled chains (inference seed 1234, temperature 0.8, top- p 0.95, 192 generated tokens) followed by majority voting. Numeric answers are canonicalized before voting. We report exact counts, Wilson 95% intervals for each accuracy, and a two-sided exact McNemar test for the paired greedy-versus-vote outcomes. Complete generations, extracted answers, seeds, settings, resumable checkpoints, and integrity tests are provided.

4 Results

4.1 Capability growth: three benchmark eras

Knowledge benchmarks saturated. The MMLU series rose from 43.9% (GPT-3, 2020) to 86.4% (GPT-4, 2023) to 91.8% (o1, late 2024), past the 89.8% estimated human-expert level. By 2024 the benchmark had lost discriminative power: four strong models clustered within 4 points²⁴, and benchmark mirrors now mark MMLU as “saturated” and exclude it from scoring¹¹. The field responded with harder knowledge tests (MMLU-Pro²⁴, Humanity’s Last Exam) and, increasingly, with agentic tasks.

Agentic coding became the frontier metric — and moved fast. Figure 2 shows SWE-bench Verified from October 2024 to July 2026. A log-linear fit on the solve odds gives $\text{odds}_{t+1\text{yr}} \approx 5.8 \times \text{odds}_t$ ($R^2 = 0.78$, $n = 14$): the odds of resolving a real GitHub issue multiplied by almost six each year. Independent evaluation puts the current frontier at 96–97%¹⁴, implying the benchmark itself will saturate within roughly a year — the same lifecycle GLUE and MMLU went through, on a harder task. Harder variants (SWE-bench Pro, Frontier-Bench, on which even the best model scores only 43–64%^{16,18}) are already taking over.

Takeaway. Each benchmark era compressed: GLUE lasted about two years, MMLU about four, and SWE-bench Verified is on track for about two. The capability frontier is real and fast, but the *measuring sticks* keep burning out — a theme that returns in Section 6.

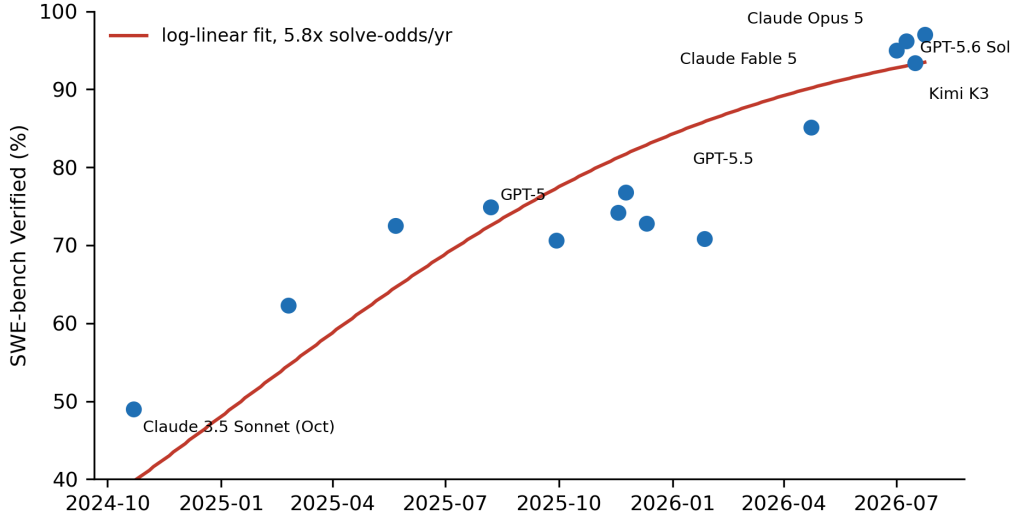


Figure 2: SWE-bench Verified, October 2024 – July 2026 (mix of vendor-reported and independent harness results; see Table 1). The log-linear fit on solve odds implies $\sim 5.8\times$ growth per year.

4.2 The collapse of the capability–cost curve

API input prices fell from GPT-3’s \$60 per million tokens (2020) to GPT-5.6 Luna’s \$1 (2026): a $\sim 60\times$ decline, compounding at roughly $2\times$ per year, *while* capability itself was compounding at a similar rate. The two effects multiply: a dollar buys orders of magnitude more capability than a dollar in 2023.

Budget tier vs. last quarter’s flagship. The sharpest illustration is internal to OpenAI’s own July 2026 launch table¹⁶. GPT-5.6 Luna (\$1/\$6 per M input/output tokens at the official standard API rate¹⁶) against GPT-5.5 (\$5/\$30), the flagship released eleven weeks earlier: across the 23 benchmarks reported for both, Luna wins 9 and loses 14 — but the wins concentrate exactly where the economic value is. On agentic and professional work (Agents’ Last Exam, GDPval-AA, SWE-bench Pro, DeepSWE, HealthBench Professional, AutomationBench, GraphWalks) Luna wins 7 of 10; it trails mainly on academic knowledge (GPQA), competition math (FrontierMath), multimodal understanding, and — sharply — long-context recall (41.3% vs. 81.5% on MRCR). Against the flagship of *one year* ago the comparison is not close: our reconstruction of overall SWE-bench Verified from Vals AI’s per-difficulty rows gives Luna 92.9% vs. GPT-5’s 74.9% vendor-reported score in August 2025^{14,25}. The practical conclusion: **capability that was flagship-class 2–12 months ago is now available at $\sim 20\%$ of the flagship token price**, with the caveat that the cheapest tier gives up long-context recall and the hardest academic reasoning.

The mid tier is dead. Our Pareto analysis of the fifteen GPT-5.6 settings reproduces and confirms Artificial Analysis’s finding^{19,20}: every Terra configuration is dominated — there is always a Luna or Sol setting that is smarter at no extra cost or equally smart and cheaper. Six of fifteen settings lie off the frontier. The economic structure of a model family is now bimodal: a volume tier and a premium tier, with the middle squeezed out.

4.3 The frontier fragmented by task

No single model leads the 2026 frontier; leadership is task-dependent (Figure 3):

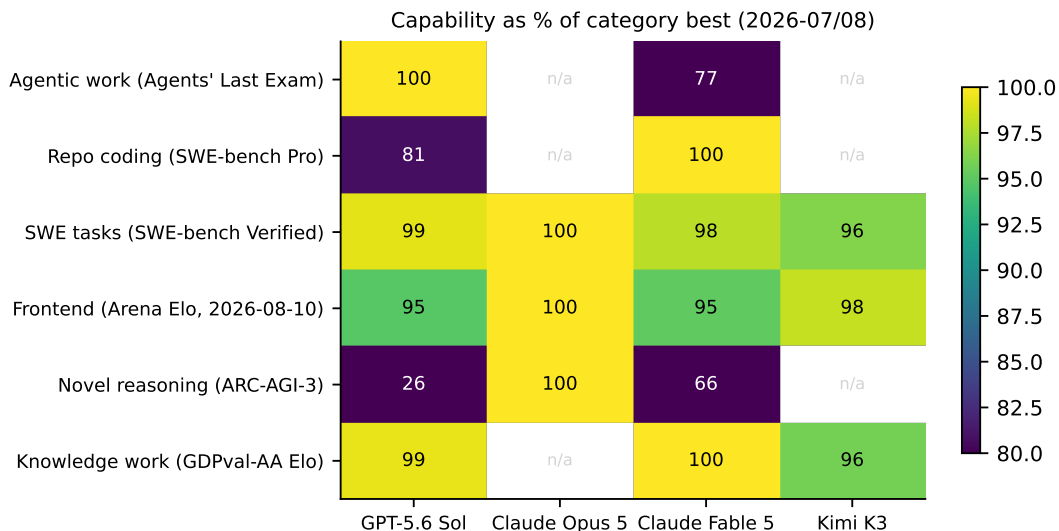


Figure 3: Capability as a percentage of the category best, July–August 2026. No column is uniformly dark: each frontier model has at least one category where another model clearly leads. Cells marked n/a lack a comparable public number — itself a symptom of vendors reporting different benchmarks.

- **Frontend and design-oriented coding: Claude Opus 5.** On the live arena.ai Web-Dev leaderboard checked August 10, Opus 5 led blind developer preference at 1,712 Elo, with open-weight Kimi K3 second at 1,682; both values were preliminary and will move as votes accumulate²¹.
- **Repository-level coding: Claude Fable 5.** Fable 5 leads SWE-bench Pro at 80.0%, 15.4 points ahead of GPT-5.6 Sol (64.6%)¹⁶.
- **Agentic terminal and long-horizon professional work: GPT-5.6 Sol.** Sol leads the Artificial Analysis Coding Agent Index (80), Terminal-Bench 2.1 (88.8%; 91.9% in multi-agent Ultra mode), and Agents' Last Exam (52.7%, +12.2 over the best non-OpenAI model)^{16,19}.
- **Novel reasoning and computer use: Claude Opus 5.** Opus 5 scored 30.2% on ARC-AGI-3 — nearly 4× the previous record of 7.8% (GPT-5.6 Sol) — plus a vendor-reported 42/42 on IMO 2026 under Anthropic's grading protocol and the lead on OSWorld 2.0 computer use^{2,3,18}.
- **Best-value agentic work: Claude Opus 5 again,** reaching within 0.5% of Fable 5's CursorBench peak at half the task cost¹⁸.

Routing recovers the optimum. If no model wins everywhere, a router should beat every individual model. On the fourteen fully-covered benchmarks, the best single model (GPT-5.6 Sol) averages 97.6% of the per-benchmark best; an oracle router averages 100% (+2.4 points). The more useful finding: a router with only *two* models — GPT-5.6 Sol and Claude Fable 5 — already captures 100% of the oracle gain on this suite, because their strengths are complementary (agentic vs. repo-level/math work). A cost-constrained router that picks the cheapest model within one point of the best matches the oracle score at 95% of its cost — the savings are modest precisely because Sol's wins are usually by wide margins (Section 5 discusses implications).

Why models diverge. Specialization is manufactured, not accidental. Moonshot optimized K3 for agentic coding with a 2.5× scaling-efficiency gain from architectural changes

Table 2: Self-consistency results (this paper’s experiment).

decoding strategy	correct / n	accuracy (%)	95% Wilson interval
greedy ($k = 1$, no sampling)	1 / 16	6.2	[1.1, 28.3]
sample-and-vote, $k = 1$	1 / 16	6.2	[1.1, 28.3]
sample-and-vote, $k = 2$	1 / 16	6.2	[1.1, 28.3]
sample-and-vote, $k = 4$	2 / 16	12.5	[3.5, 36.0]
sample-and-vote, $k = 8$	2 / 16	12.5	[3.5, 36.0]

(Kimi Delta Attention, Attention Residuals) and a training pipeline openly aimed at the agentic benchmark suite¹⁷; its 1M-token single-agent setup was deliberately tuned for long-horizon tasks like BrowseComp¹⁷. OpenAI trained GPT-5.6 for terminal and multi-agent workflows (Ultra mode natively fans out subagents)¹⁶. And the cautionary case: Opus 5’s record ARC-AGI-3 gain partially failed to transfer to Witness, an independent held-out puzzle benchmark (43.4, statistically tied with K3 and Fable 5), consistent with training on genre-adjacent data^{26,27}. Targeting a task family works — but gains measured on the targeted benchmark overstate the transferable component. We return to this in Section 6.

5 Getting more from existing models: three suggestions, tested

The analyses above suggest three actionable ways to improve effective model performance without training anything. We test each with code.

5.1 Suggestion 1: Spend test-time compute; sample and vote

Claim. For problems with verifiable answers, sampling several independent chain-of-thought solutions and majority-voting them (self-consistency²⁸) yields a materially more accurate system from the *same* model.

Experiment. We evaluate Qwen2.5-0.5B-Instruct — a half-billion parameter model that runs on a laptop CPU — on a fixed seeded sample from the committed official GSM8K test set. The paper run fails rather than silently substituting synthetic data. Two conditions: greedy decoding, and self-consistency with k samples at temperature 0.8, voted. We also report the vote at $k \in \{1, 2, 4\}$ to trace the compute–accuracy curve.

Result. In this 16-item GSM8K pilot with Qwen2.5-0.5B-Instruct, greedy decoding achieved 6.2% (1/16; 95% Wilson CI 1.1–28.3) accuracy, while sample-and-vote with $k = 8$ reached 12.5% (2/16; 95% Wilson CI 3.5–36.0) — an absolute gain of 6.2 points (paired exact McNemar $p = 1.0000$). The complete trace contains one auditable record for each paired item, including the greedy generation, all eight sampled generations, extracted answers, and prefix votes. Because this is a small pilot, its Wilson intervals are wide and the paired test should be read as uncertainty rather than a binary success threshold; the run demonstrates a reproducible mechanism and logging method, not a precise population effect. The aggregation component is expected to grow with model scale and problem difficulty in the original study²⁸. All raw generations are released with the code ([results/exp4_raw_generations.jsonl](#)). This is the small-scale shadow of the frontier pattern, where multi-sample and multi-agent inference (Sol Ultra’s 91.9% vs. 88.8% single-agent on Terminal-Bench¹⁶) buys points that no single pass can.

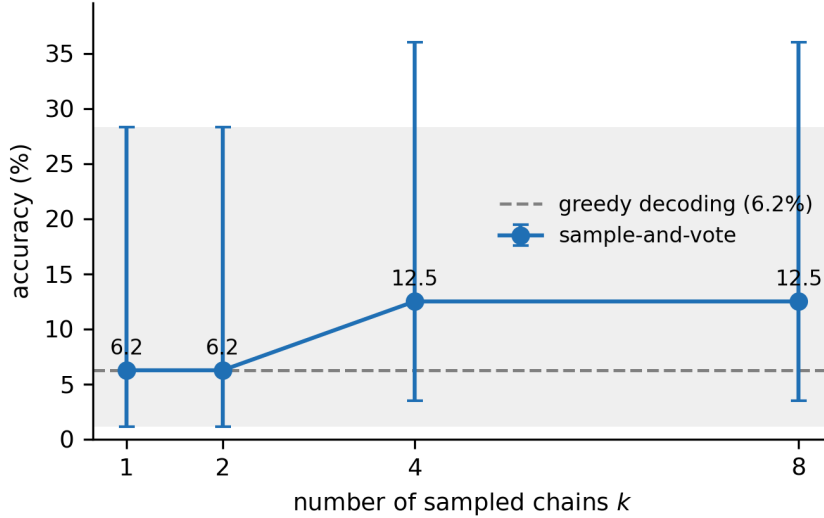


Figure 4: Accuracy of Qwen2.5-0.5B-Instruct on GSM8K as a function of sampled chains k (majority vote), versus the greedy baseline (dashed). Error bars and the gray band are Wilson 95% intervals. Data and code: [code/exp4_self_consistency.py](#).

5.2 Suggestion 2: Calibrate reasoning effort; more is not better

Claim. Cranking reasoning effort to maximum is often worse *and* more expensive.

Evidence. Two independent data points. First, Anthropic’s own score-vs-cost curves show Opus 5’s Frontier-Bench score peaks at *high* effort (44.4%) and *drops* at max (43.3%) while costing more; community re-analysis of Anthropic’s Frontier-code chart found the peak at medium effort ($\sim 53\%$)¹⁸. Second, our Pareto analysis shows each ~ 4 -point intelligence gain from Luna-max to Terra-max to Sol-max multiplies per-task cost by $2.6\times$ and $1.9\times$ respectively²⁰. A plausible mechanism is overthinking: on tasks with a direct solution path, longer chains can wander. **Recommendation:** treat effort as a per-workload hyperparameter with a measured optimum, not a quality dial to max out.

5.3 Suggestion 3: Route tasks to specialized models

Claim. Because the frontier is task-fragmented (Section 4.3), routing each workload to the right model beats committing to any single model.

Evidence. Our oracle analysis: best single model averages 97.6% of per-benchmark best across the fourteen fully-covered benchmarks; an oracle scores 100; and a two-model router (Sol + Fable 5) already captures the entire gain. Production teams are converging on this pattern — model-abstraction layers that swap models at the routing layer²⁹ — and practitioners already pair a strong planner with a cheap executor (“Sol medium for architecting, Luna high for coding”)³⁰. **Recommendation:** a small router over two complementary frontier models, with a budget tier for well-defined subtasks, dominates any single-model deployment on quality; add the budget tier for cost.

5.4 What we would tell the labs

The same analyses generate suggestions for model builders: (i) report score-vs-cost curves for every benchmark, as Anthropic now does — single numbers invite cherry-picking¹⁸; (ii) close the budget-tier long-context cliff (Luna’s 41.3% MRCR vs. Sol’s 91.5%¹⁶), the one place the

cheap model is not near-flagship; (iii) publish held-out transfer evaluations alongside targeted benchmarks (the ARC-AGI-3/Witness gap in Section 4.3); and (iv) invest in mid-context retrieval rather than context length alone — K3’s BrowseComp result suggests retrieval quality, not window size, is the binding constraint¹⁷.

6 Discussion: how models get good at specific tasks

The 2026 pattern — one frontier, several winners — is the predictable output of how frontier models are now built. We see at least four levers.

1. Domain-weighted post-training. Reinforcement learning on verifiable, domain-specific rewards (unit tests for code, proof checkers for math, task-completion signals for agents) lets a lab buy capability in a chosen vertical. Moonshot’s agentic suite results and OpenAI’s terminal/agentive lead both track heavy RL investment in those task families^{16,17}.

2. Architecture as targeting. Kimi K3’s hybrid linear attention (Kimi Delta Attention) and expert-routing changes were chosen for long-context agentic efficiency, and the payoff appeared precisely on long-horizon benchmarks (BrowseComp 91.2, state of the art at release)¹⁷.

3. Inference-time structure. GPT-5.6’s Ultra mode makes multi-agent fan-out a first-class feature, buying Terminal-Bench points (88.8% → 91.9%) at higher cost¹⁶. Our self-consistency experiment (Section 5.1) is the same lever at toy scale: capability is partly a function of inference-time orchestration.

4. Benchmark-shaped optimization (and its limits). When a benchmark becomes a target, labs optimize toward it, and two failure modes follow: the benchmark saturates (GLUE, MMLU, soon SWE-bench Verified), or gains fail to transfer off-distribution. The Witness evaluation of Opus 5 is the cleanest 2026 example: a $\sim 4\times$ record on ARC-AGI-3, but only a statistical tie with K3 and Fable 5 on held-out puzzles of the same genre, including one environment where Opus 5 stated the hidden rules before its first move — suggesting familiarity, not just reasoning^{26,27}. METR’s maintainer-review study adds a second caution from a different angle: a growing share of SWE-bench-passing patches would be rejected by human maintainers (quality, breakage, or missing the issue’s point), and the grader–maintainer gap is widening over model generations³¹. Targeting works, and that is exactly why single-number claims deserve suspicion.

Implication for evaluation. Score-versus-cost curves, held-out transfer suites, and maintainer-grade review are becoming the minimum standard of evidence. Vendors already report curves¹⁸; ARC Prize now administers runs independently with published traces^{3,27}; and METR’s maintainer study is a template for auditing what benchmark passes actually mean.

Implication for buyers. The economic shape of 2026 — a budget tier at 20% of flagship token price that matches last quarter’s flagship on most professional work, plus a fragmented frontier — means the unit of procurement is no longer “a model” but a routing policy with per-task effort settings. The 2023 question “which model is best?” has been replaced by “which combination, at which effort, for which task?”

Limitations. This study has several. (i) Mid-2026 scores mix vendor and independent evaluations; we label provenance but cannot fully deconfound harness differences (e.g., GPT-5.5 scores 85.1% vendor-reported vs. 82.6% reconstructed from Vals AI’s independent harness)^{14,25}. (ii) The SWE-bench trend fit pools vendor and independent numbers and spans only 22 months; the $5.8\times$ figure should be read as an order of magnitude. (iii) The routing oracle uses benchmark identities as a proxy for task identity; real routers must infer task type from unlabeled traffic. (iv) The self-consistency experiment uses a 0.5B model on only 16 grade-school-math items by design; its intervals are correspondingly wide, and it demonstrates the measurement and logging mechanism cheaply rather than estimating a frontier effect size. (v) The field moves weekly; figures are current as of August 10, 2026.

7 Conclusion

Eight years separate BERT’s 80.5 GLUE from Opus 5’s reported 42/42 on the 2026 IMO problems. The progression was not a smooth exponential on one axis but a sequence of paradigm changes — fine-tuning, few-shot scale, alignment, reasoning, agency — each moving capability onto a new axis and each burning out the previous era’s benchmarks. Our measurements quantify three 2026 facts. Capability in agentic coding is still compounding at roughly six-fold per year. The cost of a fixed capability level is falling even faster, to the point where a \$1/M budget tier replicates most of the professionally relevant performance of an eleven-week-old flagship. And the frontier has fragmented by task: frontend coding, repository coding, agentic terminal work, and novel reasoning each have a different leader, which makes routing — not model selection — the decision that matters. A practical way to improve a deployed system today is often to leave its weights fixed while measuring better ways to spend its inference compute.

Reproducibility. All datasets (`code/data/`), analysis scripts (`exp1–exp4`), the live experiment, unit tests (`tests/`), figure generation, and this paper’s build script are included in the accompanying archive; see `README.md`. Nothing in this paper requires proprietary access to reproduce except the live benchmark numbers, whose sources and retrieval dates are documented in `code/data_sources.md`.

References

- [1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*, 2019.
- [2] Anthropic. Claude opus 5 system card. <https://www.anthropic.com/claude-opus-5-system-card>, 2026. July 24, 2026.
- [3] ARC Prize. Claude opus 5 — ARC-AGI results. <https://arcprize.org/results/anthropic-claude-opus-5>, 2026. Independently administered results, July 24, 2026.
- [4] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv:1907.11692*, 2019.
- [5] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21, 2020.

- [6] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI Technical Report*, 2019.
- [7] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *NeurIPS*, 2020.
- [8] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask understanding. *ICLR*, 2021.
- [9] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *NeurIPS*, 2022.
- [10] OpenAI. GPT-4 technical report. *arXiv:2303.08774*, 2023.
- [11] BenchLM. MMLU leaderboard & scores. <https://benchlm.ai/benchmarks/mmlu>, 2026. Retrieved August 7, 2026; benchmark marked saturated/display-only.
- [12] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? *ICLR*, 2024.
- [13] OpenAI. Introducing SWE-bench verified. <https://openai.com/index/introducing-swe-bench-verified/>, 2024.
- [14] Vals AI. SWE-bench verified — independent evaluation results. <https://vals.ai/benchmarks/swebench>, 2026. Retrieved August 6, 2026.
- [15] Anthropic. Claude fable 5 and claude mythos 5. <https://www.anthropic.com/news/claude-fable-5-mythos-5>, 2026. Official launch post, June 9, 2026; availability updated July 1, 2026.
- [16] OpenAI. GPT-5.6: Frontier intelligence that scales with your ambition. <https://openai.com/index/gpt-5-6/>, 2026. Launch tables, July 9, 2026; retrieved August 2026.
- [17] Moonshot AI. Kimi k3: Open frontier intelligence. <https://github.com/MoonshotAI/Kimi-K3>, 2026. Official repository and technical report, July 16, 2026.
- [18] Anthropic. Introducing claude opus 5. <https://www.anthropic.com/news/claude-opus-5>, 2026. Official launch post, July 24, 2026.
- [19] Artificial Analysis. GPT-5.6 benchmarks across intelligence, speed and cost. <https://artificialanalysis.ai/articles/gpt-5-6-has-landed>, 2026. July 9, 2026.
- [20] Igor Warzocha. Comparison for all 5.6 models (artificial analysis anchor measurements). <https://gist.github.com/IgorWarzocha/60bfd11731f15cf8802f0b6e80d47ac7>, 2026. July 11, 2026.
- [21] Arena AI. Code arena: Webdev overall leaderboard. <https://arena.ai/leaderboard/code/webdev/overall>, 2026. Live blind-preference leaderboard, retrieved August 10, 2026.
- [22] DeployBase. Cost per token over time: How LLM API pricing has dropped. <https://deploybase.ai/articles/cost-per-token-over-time-how-llm-api-pricing-has-dropped>, 2026. March 4, 2026.

- [23] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Rei Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv:2110.14168*, 2021.
- [24] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, et al. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. *NeurIPS Datasets and Benchmarks*, 2024.
- [25] LocalAIMaster. GPT-5.5 review: ChatGPT default, tested. <https://localaimaster.com/models/gpt-5-5>, 2026. May 9, 2026.
- [26] The Decoder. Anthropic’s opus 5 blows past fable 5 and gpt-5.6 sol on the benchmark designed to measure real intelligence. <https://the-decoder.com/anthropics-opus-5-blows-past-fable-5-and-gpt-5-6-sol-on-the-benchmark-designed-to-measure-real-intelligence/>, 2026. July 26, 2026; includes Witness benchmark results.
- [27] TechTimes. Claude opus 5 took ARC-AGI-3 record with an equation no AI had written before. <https://www.techtimes.com/articles/321661/20260727/>, 2026. July 27, 2026.
- [28] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. *ICLR*, 2023.
- [29] Yash Thakker. GPT-5.6: Rolling out july 9 — sol, terra, luna. <https://explainx.ai/blog/gpt-5-6-release-date-features-benchmarks-2026>, 2026. June–July 2026.
- [30] Hacker News. Advancing the price-performance frontier with GPT-5.6 — discussion. <https://news.ycombinator.com/item?id=49112867>, 2026. July 30, 2026; community benchmark comparisons.
- [31] METR. Many SWE-bench-passing PRs would not be merged into main. <https://metr.org/notes/2026-03-10-many-swe-bench-passing-prs-would-not-be-merged-into-main/>, 2026. March 10, 2026.